THE HOOK

Next-Generation Deception Infrastructure

Full Technical Architecture & MVP Blueprint

CONFIDENTIAL — TECHNICAL DESIGN DOCUMENT

Classification: Internal / Investor Use Only

Version 1.0 | April 2025

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

EXECUTIVE SUMMARY

THE HOOK is a next-generation deception infrastructure platform that inverts the economics of cyberattacks. Rather than attempting to block every intrusion attempt — a fundamentally reactive strategy — THE HOOK deploys a field of realistic fake infrastructure that lures, traps, and silently studies attackers while keeping production systems safe.

This document presents the complete technical architecture, MVP implementation roadmap, and commercialisation strategy for THE HOOK platform. It is designed to serve as both a technical blueprint for a development team and a strategic overview for investors and enterprise customers.

CORE THESIS

Every attacker who enters THE HOOK's deception environment is an asset — a source of intelligence. Their tools, techniques, and objectives are captured, analysed, and converted into defensive value for the entire customer base.

Platform Attribute	Value
Primary Defence Paradigm	Active Deception / Threat Intelligence
Target Customers	Mid-market & enterprise, MSSPs
Deployment Model	SaaS, on-premise, hybrid
MVP Development Timeline	12–16 weeks (small team)
Target ARR at Year 2	€2M–€5M
Core IP Differentiator	Adaptive attacker engagement engine

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 01 — SYSTEM OVERVIEW

1.1 Why Traditional Defences Fail

The dominant model in enterprise cybersecurity is perimeter defence: firewalls, intrusion detection systems, endpoint protection, and SIEM platforms. These tools share a fundamental structural weakness — they are reactive. They wait for known threat signatures, suspicious network patterns, or anomaly thresholds to trigger alerts. Against modern adversaries, this is systematically insufficient for three reasons.

- Asymmetric information advantage: Attackers operate with full knowledge of their own tools, timing, and objectives. Defenders must identify a threat from noise across millions of events per day. The attacker needs to succeed once; the defender must succeed every time.
- Zero-day and living-off-the-land attacks: Sophisticated threat actors increasingly use legitimate system tools and previously unknown vulnerabilities. Signature-based detection is blind to these techniques by design.
- Alert fatigue and analyst capacity: Enterprise SOCs routinely process hundreds of thousands of alerts per day, the vast majority of which are false positives. Genuine intrusions frequently go undetected for weeks or months — the 2024 Mandiant M-Trends report places median dwell time at 10 days globally, but in many sectors the figure is far higher.

The consequence is a security model in which defenders are perpetually behind the adversary, operating with incomplete information, under resource constraints, and forced to make high-stakes triage decisions with insufficient context.

1.2 How Deception Infrastructure Changes the Strategy

Deception technology, also known as active defence or moving target defence, reverses this dynamic. Instead of attempting to identify every malicious action within a real environment, deception infrastructure populates the network with plausible-looking fake assets — servers, databases, credentials, endpoints, and cloud storage — that have no legitimate business function. Any interaction with these assets is definitionally suspicious.

KEY INSIGHT	In a deception environment, there are no false positives. If something touches a honeypot, it is either an attacker or a misconfigured internal system. The signal-to-noise ratio approaches 1:1 — the inverse of a conventional SIEM deployment.

THE HOOK extends classical honeypot theory into a fully instrumented platform. It does not merely detect intrusions — it actively engages attackers, maintains their presence inside the fake environment, and extracts maximum intelligence before they realise they are inside a trap.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

1.3 The Value of Attacker Intelligence

The intelligence generated inside THE HOOK's deception environment has compounding value across three dimensions:

- Tactical intelligence: What specific vulnerability did the attacker exploit to gain initial access? What lateral movement techniques did they use? What credentials did they attempt? This data is immediately actionable for patching and hardening real infrastructure.
- Strategic intelligence: What is this threat actor interested in? Financial data, source code, user records, operational systems? Understanding attacker objectives helps organisations prioritise their defences correctly.
- Threat feed intelligence: IP addresses, domain names, malware samples, and attack toolkits captured inside THE HOOK can be shared (with appropriate anonymisation) across the customer base and with broader threat intelligence ecosystems such as MISP, OpenCTI, and commercial threat feeds.

Over time, as THE HOOK is deployed across a larger customer base, the intelligence network effect compounds: each new deployment contributes signals that improve detection and response for all other customers.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

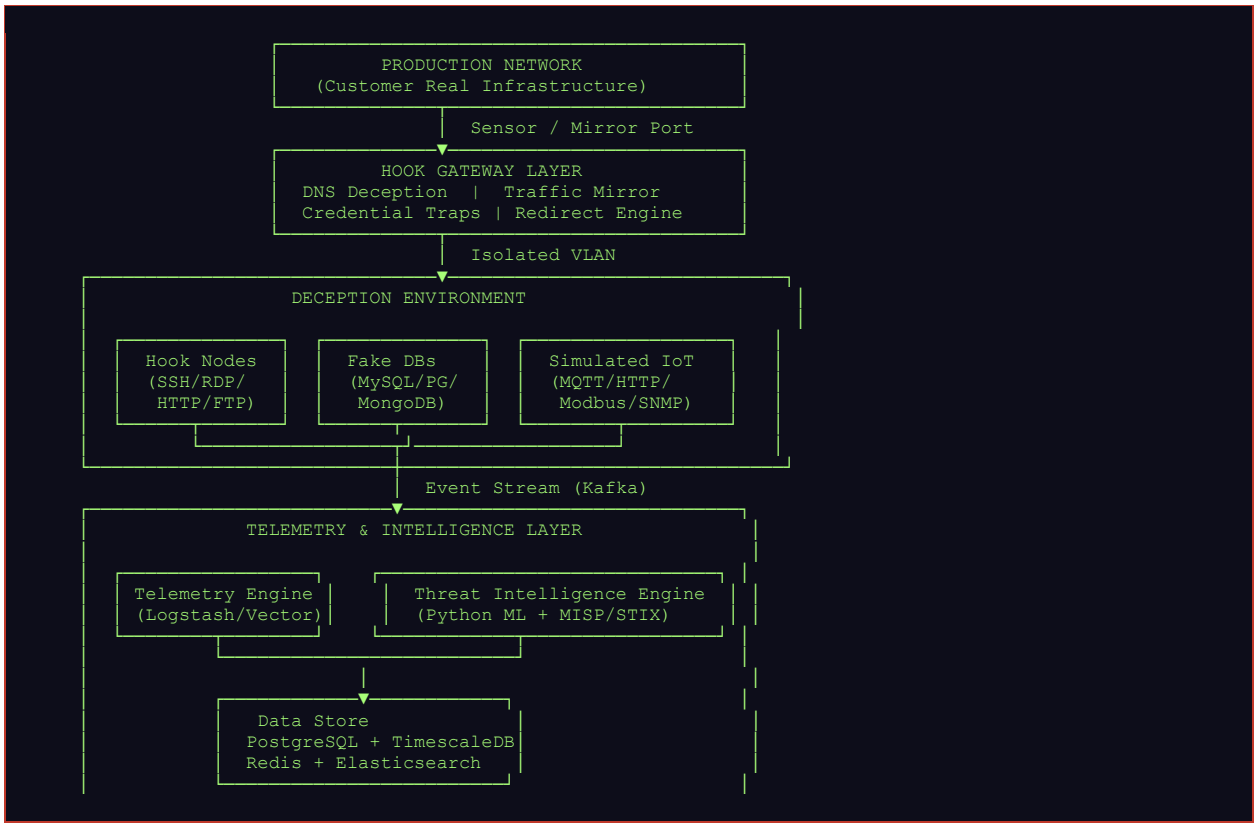
© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 02 — HIGH-LEVEL ARCHITECTURE

2.1 Architectural Principles

- Complete isolation: The deception environment must be physically and logically isolated from production infrastructure. A breach of the deception layer must carry zero risk to real systems.
- Realistic fidelity: Fake assets must be indistinguishable from real ones to a determined attacker using automated scanners and manual reconnaissance techniques.
- Full observability: Every packet, every keystroke, every file access inside the deception environment is captured, timestamped, and made available for analysis.
- Operational simplicity: The platform must be deployable and manageable by a security team without deep deception-technology expertise.

2.2 Architecture Diagram



CONFIDENTIAL | Technical Architecture v1.0 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

```
└──────────────────────────────────────────────────────────┘
                              │  REST / WebSocket API
                        ┌─────────▼──────────────────────────┐
                        │          DASHBOARD & ALERTING      │
                        │  React UI  |  PagerDuty  |  Slack  │
                        │  SIEM Integration (Splunk / Sentinel) │
                        └────────────────────────────────────┘
```

The architecture is divided into four horizontal layers with strict one-way data flow from the production boundary down to the dashboard. Attackers enter at the Gateway Layer, interact exclusively with the Deception Environment, generate events that flow into the Intelligence Layer, and are ultimately presented as structured intelligence in the Dashboard.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 03 — CORE COMPONENTS

3.1 Hook Nodes

Hook Nodes are the primary interactive deception endpoints — the fake servers that attackers encounter, probe, and attempt to exploit. Each Hook Node is a containerised service running inside an isolated Kubernetes namespace, emulating a specific technology profile (Linux web server, Windows domain controller, VoIP gateway, industrial control system, etc.).

- Service emulation: Each node emits correct protocol banners, serves realistic HTTP responses, accepts SSH connections, responds to SNMP polls, and generally behaves like a production system.
- Personality profiles: Nodes are assigned 'personalities' — a combination of OS fingerprint, software stack, configured hostname, and network topology position — that are consistent with the target organisation's real infrastructure.
- Interaction depth: Nodes operate at three levels of interaction depth: low (banner-only), medium (simulated shell or web application), and high (full filesystem simulation with fake sensitive data).
- Auto-scaling: The Kubernetes orchestrator automatically spawns and destroys Hook Nodes based on demand and configured deception coverage policies.

3.2 Deception Services

Deception Services are the protocol-level implementations embedded within Hook Nodes. They differ from full emulation environments in that they are lightweight, purpose-built listeners designed to maximise intelligence capture with minimal resource consumption.

Service	Protocol	Intelligence Captured
FakeSSH	SSH/22	Credentials, public keys, commands
FakeHTTP/S	HTTP/443	Requests, payloads, user-agents, exploits
FakeSMB	SMB/445	Share access, credential hashes, file operations
FakeRDP	RDP/3389	Credential attempts, session metadata
FakeFTP	FTP/21	Login attempts, file transfer patterns
FakeSQL	MySQL/3306, PG/5432	Query patterns, injection

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

		attempts
FakeMQTT	MQTT/1883	IoT command injection, topic enumeration
FakeModbus	Modbus/502	ICS/SCADA enumeration and manipulation
FakeSMTP	SMTP/25	Spam relay attempts, credential stuffing
FakeRedis	Redis/6379	Unauthorised command execution patterns

3.3 Telemetry Collectors

Telemetry Collectors are lightweight agents deployed at every data capture point across the deception environment. They gather raw event data — network packets, process events, file system operations, system calls — and normalise it into a structured schema before forwarding it to the central event stream.

- Network capture: libpcap-based packet capture on all Hook Node interfaces, providing full packet inspection for off-line analysis.
- Session recording: Full keystroke and session recording for interactive protocols (SSH, RDP, Telnet). Every command typed by an attacker is preserved.
- Syscall tracing: In high-interaction environments, eBPF probes capture system calls, giving visibility into malware behaviour at the kernel level.
- Event normalisation: All events are normalised to a common schema (ECS-compatible) before ingestion, enabling cross-protocol correlation.

3.4 Threat Intelligence Engine

The Threat Intelligence Engine (TIE) is the analytical core of THE HOOK. It consumes the normalised event stream and performs real-time and batch analysis to extract structured threat intelligence from raw attacker behaviour.

- IP reputation enrichment: Every attacker IP is enriched in real-time against commercial and open-source threat feeds (VirusTotal, AbuseIPDB, Shodan, internal reputation scores).
- TTP mapping: Attack sequences are automatically mapped to MITRE ATT&CK tactics, techniques, and procedures (TTPs) using a rule engine and ML classifier.
- Malware detonation pipeline: Uploaded files are automatically forwarded to a sandboxed malware analysis environment (CAPEv2 integration) for dynamic analysis.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

- Behavioural fingerprinting: Attacker tool signatures, timing patterns, and command vocabularies are used to build attacker profiles and cluster related campaigns.

3.5 Monitoring Dashboard

The Dashboard is the primary interface through which security analysts and platform administrators interact with THE HOOK. It provides real-time visibility into active attacker sessions, historical intelligence, alert management, and platform health.

3.6 Alerting and Integration Layer

The Alerting Layer is responsible for converting intelligence events into actionable notifications and sharing data with external security platforms. It supports webhook delivery, email, PagerDuty, Slack, Microsoft Teams, and SIEM integration via CEF/Syslog forwarding and native connectors for Splunk, Microsoft Sentinel, and IBM QRadar.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 04 — DECEPTION INFRASTRUCTURE

4.1 Design Philosophy: The Art of the Convincing Fake

The fundamental challenge of deception infrastructure is fidelity. A honeypot that is trivially identifiable as fake provides no intelligence value — sophisticated attackers will simply avoid it, or worse, use it to fingerprint the defender's capabilities. THE HOOK addresses this challenge through a layered realism model.

PRINCIPLE	A Hook Node does not need to be a perfect replica of a production system — it needs to be convincing enough that an attacker will spend time inside it before suspecting deception. Every minute of attacker time inside the deception environment is intelligence collected and real systems protected.

4.2 Fake Server Infrastructure

Fake servers are deployed in three configurations, calibrated to the threat model and network topology of each customer deployment:

- Breadcrumb servers (low interaction): These are minimal-footprint services that respond to scanning and initial enumeration, capturing attacker reconnaissance data. They present realistic banners and respond to basic protocol handshakes but do not maintain persistent interactive sessions. Deployed in high volume across customer network ranges.
- Engagement servers (medium interaction): These servers accept interactive connections and present a simulated environment — a plausible SSH session with a limited command set, a web application with navigable pages, or a file server with browsable share structure. Attackers can explore but will encounter resistance if they attempt deep exploitation.
- Deep capture servers (high interaction): Full-emulation environments running real operating systems inside hardened containers or VMs, with instrumented file systems, fake sensitive data, and real (sandboxed) services. Designed to capture sophisticated attackers who would detect a medium-interaction honeypot.
- OS Fingerprint management: Nmap OS detection is defeated by configuring IP stack parameters (TCP window size, TTL, MSS, IP ID sequence) to match the target OS profile. P0f passive fingerprinting is similarly addressed through careful protocol implementation.

4.3 Fake Databases

Database honeypots are among the highest-value deception assets because attackers who reach a database are typically in a late stage of their kill chain and are actively seeking to exfiltrate valuable data.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

- MySQL/PostgreSQL honeypots: Accept real database connections, present a plausible schema (fake HR records, financial tables, customer data with realistic-but-synthetic PII), and log all queries. Exfiltration attempts are captured and the data extracted (fake data) is tracked.
- MongoDB/Elasticsearch honeypots: Emulate commonly misconfigured NoSQL instances — these are frequently targeted in automated mass-exploitation campaigns. The honeypot logs the attacker's IP, tooling signatures, and the specific collections they access.
- Redis honeypots: Emulate unauthenticated Redis instances, a common misconfiguration in cloud environments. Attackers frequently attempt to use Redis for persistence (cron-based reverse shell installation) — these attempts are captured in full.

4.4 Simulated IoT and OT Devices

Industrial Control System (ICS) and Internet of Things (IoT) environments represent an increasingly targeted attack surface. THE HOOK provides a library of pre-built OT/IoT honeypot profiles:

- Smart building systems: BACnet, KNX, and Modbus honeypots simulating HVAC, access control, and lighting systems.
- Industrial PLCs: Siemens S7, Allen-Bradley, and generic Modbus TCP honeypots emulating programmable logic controllers.
- Network infrastructure: Fake Cisco IOS, Juniper JunOS, and Fortinet FortiGate devices responding to SNMP polls and accepting SSH/Telnet sessions.
- IP cameras and NVRs: RTSP/ONVIF honeypots emulating CCTV systems, commonly targeted in botnet recruitment campaigns.

4.5 Fake Cloud Storage

Cloud storage misconfigurations (exposed S3 buckets, Azure Blob containers, GCS buckets) are among the most commonly exploited vulnerabilities in modern cloud environments. THE HOOK deploys fake cloud storage endpoints that mimic the API interfaces of major providers:

- S3 API emulation: A fake S3 endpoint seeded with plausible-but-synthetic data (fake backups, configuration files with dummy credentials, financial exports). Any access is immediately alerted.
- Credential trap files: Files planted in fake storage endpoints contain fake AWS access keys and database credentials. If an attacker attempts to use these credentials elsewhere, the attempt is detected via a canary token callback.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

4.6 Simulated Internal Company Networks

The most sophisticated deception deployments present attackers with an entire simulated internal network — a ghost of the organisation's real infrastructure. This is THE HOOK's most powerful capability and the primary differentiator from simple honeypot solutions.

- Active Directory forest: A fake AD domain with realistic OU structure, user accounts, group policies, and service accounts. Attackers who gain a foothold will attempt to enumerate and move laterally through this fake domain, generating rich intelligence.
- Internal application stack: Fake internal applications (HR portal, ticketing system, source code repository, wiki) with realistic-looking content and plausible authentication flows.
- Email infrastructure: A fake Exchange/mail server that accepts connections and logs all commands. Spear-phishing frameworks and email harvesting tools will interact with it during reconnaissance.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 05 — ATTACK REDIRECTION

5.1 Redirection Architecture Overview

Attack redirection is the mechanism by which THE HOOK ensures that attackers who are targeting real infrastructure are quietly redirected into the deception environment without their awareness. This is distinct from simple honeypot deployment — redirection requires active intervention in the network traffic flow.

5.2 DNS Deception

DNS-based redirection is the highest-leverage technique available to THE HOOK platform. By manipulating DNS responses — either at the internal resolver level or via a DNS firewall — the platform can redirect entire categories of traffic:

- Internal DNS poisoning: For known malicious domains (identified through threat intelligence feeds), THE HOOK's DNS resolver returns an IP address pointing to a Hook Node. The attacker's malware attempts to beacon home and instead connects to a deception server.
- Typosquat interception: Common typographical variants of the organisation's internal domain names are registered and resolved to Hook Nodes. Attackers who enumerate DNS or attempt to access internal services via guessed hostnames land in the deception environment.
- Sinkholing with intelligence capture: Rather than simply NXDOMAINing malicious domain resolutions (standard sinkhole behaviour), THE HOOK resolves them to interactive services, enabling the attacker to proceed while being observed.

5.3 Network Segmentation and VLAN Injection

At the network layer, THE HOOK Gateway sits inline or passively on a mirrored port between the internet-facing DMZ and the internal production network. When a connection attempt targeting a non-existent internal IP address is detected, the Gateway intercepts it and redirects it to an appropriate Hook Node.

- Dark space monitoring: IP addresses within the organisation's allocated ranges that are not assigned to production systems are monitored. Any traffic to these 'dark' IPs is definitionally suspicious and is redirected to deception infrastructure.
- VLAN-level isolation: The deception environment occupies a physically segregated VLAN with no routing path to production systems. The gateway enforces this isolation at the hardware level.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

5.4 Traffic Mirroring

Traffic mirroring (SPAN or TAP-based) allows THE HOOK to observe all network traffic passively without inserting itself into the traffic flow. This provides zero-impact monitoring of production traffic and enables detection of attacker reconnaissance before they ever interact with deception assets.

- Passive fingerprinting: mirrored traffic is processed through p0f and Suricata to identify attacker tools and techniques based on traffic signatures alone.
- Trigger-based redirection: When passive monitoring detects a scanning pattern or exploitation attempt, the platform can dynamically insert a BGP route or ARP entry to redirect the attacker's next connection to a deception asset.

5.5 Credential Traps

Credential traps — sometimes called 'honeytoken credentials' or 'canary tokens' — are fake credentials planted in locations where a legitimate user would never use them, but an attacker performing credential harvesting would collect and attempt to use.

- Browser credential injection: Fake saved passwords planted in browser profiles on compromised-looking endpoints. If used, the login attempt triggers an immediate high-priority alert.
- Configuration file credentials: Database connection strings and API keys embedded in fake configuration files, source code repositories, and environment variable dumps. Canary token callbacks provide real-time notification of any usage attempt.
- Active Directory service accounts: Fake service accounts with 'juicy' names (svc_backup, admin_legacy, dbadmin) and weak passwords that are never used in production. Any authentication event using these accounts is definitionally malicious.
- AWS IAM canary keys: Fake AWS access key pairs with CloudTrail monitoring. Any API call made with these keys triggers an immediate alert, regardless of where in the world the call originates.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 06 — ATTACKER INTERACTION SYSTEM

6.1 The Engagement Problem

A deception environment that triggers an immediate alert the moment an attacker touches it provides limited intelligence value. The attacker has revealed only that they scanned a particular IP — information of modest utility. The goal of the Attacker Interaction System is to maximise attacker dwell time inside the deception environment, extracting the maximum possible intelligence before the attacker either achieves their (fake) objective or detects the deception.

6.2 Fake Login Systems

Login interfaces are the primary interaction point for most attackers. THE HOOK implements fake login systems for SSH, RDP, web applications, VPNs, and database management consoles. These systems are carefully designed to balance two requirements:

- Enough friction to appear real: Instant login acceptance is a honeypot tell. Real systems have latency. THE HOOK introduces configurable authentication delays and periodic failures to mimic real server behaviour.
- Eventual access to maintain engagement: After a configurable number of attempts, the fake login system accepts the attacker's credentials — regardless of what they are — and admits them to the simulated environment. This ensures persistent engagement even against credential-stuffing attacks using large password lists.
- Multi-factor authentication simulation: For web application honeypots, THE HOOK simulates MFA challenges (TOTP prompt, SMS code simulation) to identify attackers who have pre-compromised MFA devices or tokens.

6.3 Simulated File Systems

Once inside a Hook Node, attackers will explore the file system looking for valuable data. THE HOOK provides a dynamically generated fake file system that presents a realistic directory structure, populated with synthetic data crafted to match the organisation's profile.

- Content generation: Files are generated using template-based synthesis — fake invoices in PDF format, Excel spreadsheets with financial data, Word documents with HR records, source code files in the organisation's apparent technology stack.
- File access tracking: Every file open, read, copy, or deletion operation is logged. The platform tracks exactly which fake files the attacker accessed, providing insight into their specific objectives.
- Delayed exfiltration alerting: Rather than alerting immediately when an attacker accesses fake sensitive files, THE HOOK can be configured to wait until an exfiltration attempt is

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

observed, ensuring that the attacker's exfiltration tooling and destination infrastructure are also captured.

6.4 Fake Credentials Embedded in the Environment

The fake environment is seeded with credentials at every level — environment variables, configuration files, bash history, browser saved passwords, and network shares. These credentials appear to grant access to further systems (further deception infrastructure), creating a breadcrumb trail that leads attackers deeper into the deception environment and away from real systems.

6.5 Adaptive Environment Engine

The Adaptive Environment Engine (AEE) is THE HOOK's most sophisticated component. It continuously monitors attacker behaviour within the deception environment and dynamically adjusts the environment to maintain engagement and maximise intelligence capture.

- Behaviour-responsive content: If an attacker's activity suggests they are looking for financial data, the AEE populates accessible directories with financial documents. If they appear to be hunting for source code, source code repositories become visible.
- Attacker pacing: If the attacker appears to be moving quickly (suggesting automated tooling), the environment introduces latency and access throttling to slow them down and generate richer telemetry. If they appear to be dormant, the environment generates realistic background activity (cron job outputs, log file updates) to maintain the appearance of a live system.
- Persistence simulation: When attackers install persistence mechanisms (cron jobs, systemd services, registry run keys), the environment accepts and logs the payload while presenting fake evidence of successful persistence. The attacker believes they have successfully established a foothold, which encourages continued engagement.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 07 — THREAT INTELLIGENCE COLLECTION

7.1 Telemetry Collection Architecture

Telemetry collection in THE HOOK is designed around a streaming event architecture. All events generated within the deception environment are published to a central Apache Kafka cluster in real time. Kafka provides the durability, throughput, and consumer group semantics required to support multiple downstream consumers without data loss or inter-consumer interference.

```
Event flow: Hook Node → Telemetry Collector Agent → Kafka Topic → TIE Consumers
```

Each event is published as a structured JSON document conforming to the Elastic Common Schema (ECS), enabling direct ingestion into Elasticsearch and compatibility with existing SIEM platforms. The schema is extended with THE HOOK-specific fields for deception context, attacker session identifiers, and intelligence classification.

7.2 IP Tracking and Attribution

IP attribution is the first step in contextualising any attacker interaction. The platform performs real-time enrichment for every source IP observed in the deception environment:

- Geolocation: MaxMind GeoIP2 provides city-level geolocation with ASN and organisation attribution.
- Reputation scoring: AbuseIPDB, VirusTotal IP reports, and Shodan data are queried to establish a prior reputation score for the IP. This score is used to prioritise alerts and weight intelligence outputs.
- Historical correlation: The IP is checked against THE HOOK's internal threat database to determine whether it has been observed in previous campaigns across the customer base.
- TOR/VPN/Proxy detection: Known TOR exit nodes, commercial VPN egress IPs, and residential proxy networks are tagged to distinguish infrastructure attribution from attacker attribution.

7.3 Attack Pattern Recognition

Attack patterns are identified through a combination of deterministic rule matching and machine learning classification:

- Signature matching: Known attack tool signatures (Nmap, Metasploit, Cobalt Strike, BloodHound, Mimikatz, etc.) are matched against network traffic and command patterns using Suricata rules and YARA signatures.
- MITRE ATT&CK mapping: A rule engine maps observed TTPs to the MITRE ATT&CK framework in real time. The platform maintains a per-session ATT&CK heat map that tracks which techniques have been observed in the current attacker session.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

- Campaign clustering: ML clustering algorithms group related attacker sessions based on shared infrastructure, tool signatures, timing patterns, and TTP combinations. This enables identification of coordinated campaigns across multiple customer deployments.
- Sequence analysis: LSTM-based sequence models identify unusual command execution patterns that deviate from normal attacker behaviour baselines, flagging potentially novel techniques for analyst review.

7.4 Malware Capture and Analysis

- Static analysis: Uploaded files are immediately submitted to a YARA rule engine and a PE analyser for static signature and structural analysis.
- Dynamic sandbox: Files are automatically detonated in a CAPEv2 sandbox environment, generating a behavioural report including network connections, registry modifications, file system changes, and API call sequences.
- STIX/TAXII export: Malware indicators are automatically packaged as STIX 2.1 bundles and made available via a TAXII 2.1 server for sharing with external threat intelligence platforms.

7.5 Intelligence Storage Architecture

Data Type	Storage Technology
Raw event stream	Apache Kafka (7-day retention)
Structured events / search	Elasticsearch 8.x (hot/warm/cold tiering)
Time-series metrics	PostgreSQL + TimescaleDB
Session recordings	Object storage (S3/MinIO) + AES-256 encryption
Attacker profiles	PostgreSQL (relational)
Threat intelligence	MISP + OpenCTI (graph store)
Real-time state	Redis 7.x (sorted sets + streams)
Malware samples	Encrypted object store (isolated)

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 08 — MONITORING DASHBOARD

8.1 Dashboard Architecture

The Dashboard is a single-page React application served from a dedicated frontend cluster, communicating with the backend via a REST API and WebSocket connections for real-time event streaming. Authentication is handled via OAuth2/OIDC with support for SAML federation for enterprise SSO integration.

8.2 Core Views

Attacker World Map

The primary landing screen presents a real-time world map overlaid with attacker origin geolocations. Active sessions are displayed as animated attack vectors on the map. Each dot is colour-coded by threat severity (low / medium / high / critical), and clicking on a location opens a sidebar with detailed session information, enrichment data, and a timeline of attacker actions.

Live Session Monitor

The session monitor provides a real-time view of all active attacker sessions across the deception environment. Each session card shows: source IP and geolocation, target Hook Node, active protocol, session duration, current ATT&CK stage, number of credentials attempted, and a live command stream for interactive sessions.

Threat Intelligence Feed

A scrolling intelligence feed displays newly generated threat indicators in real time: new IP addresses, malware hashes, attack patterns, and cross-customer campaign observations. Analysts can click on any indicator to open a detailed intelligence report and take actions such as blocking, exporting to SIEM, or sharing to a threat feed.

Analytics and Reporting

The analytics view provides trend analysis over configurable time ranges: attack volume by protocol, top source ASNs, TTP distribution heatmaps aligned to MITRE ATT&CK, campaign timelines, and per-node engagement statistics. All charts are exportable as PDF reports or raw data in CSV/JSON format.

Deception Coverage Map

A network topology view showing the customer's deception footprint — all deployed Hook Nodes with their current status, interaction level configuration, and recent activity. Analysts can deploy, reconfigure, or decommission nodes directly from this view.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

Alert Management

The alert queue presents all platform-generated alerts sorted by severity and time. Alerts include: first interaction with a Hook Node, use of a canary credential, malware upload detection, high-value command execution (privilege escalation, data exfiltration attempt), and threat intelligence matches. Each alert supports acknowledge, assign, comment, escalate, and close workflows.

8.3 Integration Panels

The dashboard includes integration management panels for configuring SIEM forwarding, threat feed subscriptions, ticketing system integration (Jira, ServiceNow), and notification channel configuration. All integrations are managed through a unified connector framework and support webhook and API-based delivery.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 09 — MVP IMPLEMENTATION PLAN

9.1 MVP Scope Definition

The MVP is defined as the minimum set of capabilities required to: deploy a convincing deception environment, capture and store attacker telemetry, present intelligence through a usable dashboard, and demonstrate clear value to a pilot customer. Non-essential features are deferred to V1.1 and V2.0 releases.

MVP PHILOSOPHY

The MVP must be deployable on a single Kubernetes cluster, configurable by a security engineer without custom development, and capable of generating a compelling live demonstration within 30 minutes of deployment.

9.2 MVP Components — Must Have

Component	MVP Implementation	Full Version
Hook Nodes	3 pre-built profiles (SSH, HTTP, MySQL)	20+ profiles, custom builder
Deception Services	SSH, HTTP/S, MySQL emulation	Full protocol library
Telemetry Collection	JSON log streaming to Elasticsearch	Kafka stream, eBPF capture
Dashboard	Basic React UI, session list, alert feed	Full analytics, ATT&CK heatmap
Alerting	Email + Slack webhook	Full integration library
Deployment	Docker Compose (single host)	Kubernetes Helm chart
TIE (basic)	IP enrichment, YARA matching	ML classification, campaign clustering

9.3 Development Timeline

Phase	Duration	Deliverables
Phase 1 — Foundation	Weeks 1–4	Repo structure, Docker Compose env, 3 Hook Node profiles, log pipeline to

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

		Elasticsearch
Phase 2 — Intelligence	Weeks 5–8	TIE MVP (IP enrichment, YARA), PostgreSQL schema, basic REST API
Phase 3 — Dashboard	Weeks 9–12	React dashboard (map, sessions, alerts), WebSocket real-time feed, email alerting
Phase 4 — Hardening	Weeks 13–16	Security review, demo environment, documentation, pilot deployment prep

9.4 Team Requirements

- 1× Backend engineer (Python/Go) — Hook Node emulation, TIE, API
- 1× Infrastructure engineer (Kubernetes, Docker, networking) — deployment, pipeline
- 1× Frontend engineer (React, D3.js) — dashboard UI
- 0.5× Security specialist — deception strategy, profile creation, threat intelligence

Total: 3.5 FTE for 16 weeks. Estimated cost at market rates: €140,000–€200,000 including tooling and infrastructure.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 10 — GITHUB REPOSITORY STRUCTURE

10.1 Monorepo Architecture

THE HOOK is structured as a monorepo managed with Nx or Turborepo, enabling coordinated versioning and cross-package dependency management while maintaining clear module boundaries.

```
the-hook/
├── packages/
│   ├── hook-core/              # Core orchestration engine
│   │   ├── src/
│   │   │   ├── node_manager/   # Hook Node lifecycle (spawn/destroy/health)
│   │   │   ├── session_tracker/ # Active session state management
│   │   │   ├── event_router/   # Event routing to Kafka topics
│   │   │   └── config/         # Platform configuration schema
│   │   └── Dockerfile
│   │
│   ├── deception-engine/       # Protocol emulation services
│   │   ├── protocols/
│   │   │   ├── ssh/            # Paramiko-based SSH honeypot
│   │   │   ├── http/           # Flask-based HTTP/S honeypot
│   │   │   ├── ftp/            # FTP honeypot (Python ftplib)
│   │   │   ├── mysql/          # MySQL protocol emulator
│   │   │   ├── rdp/            # RDP service fingerprint emulator
│   │   │   ├── smb/            # SMB share emulator (Impacket)
│   │   │   ├── mqtt/           # MQTT broker honeypot
│   │   │   └── modbus/         # Modbus TCP ICS emulator
│   │   ├── profiles/           # Node personality YAML configs
│   │   ├── filesystem/         # Fake filesystem generator
│   │   └── adaptive/           # Adaptive Environment Engine
│   │
│   ├── telemetry-collector/    # Event capture and normalisation
│   │   ├── agents/
│   │   │   ├── pcap_agent/     # libpcap-based network capture
│   │   │   ├── session_agent/  # Interactive session recorder
│   │   │   └── syscall_agent/  # eBPF probe (Linux kernel 5.8+)
│   │   ├── normalizer/         # ECS schema normalisation
│   │   └── kafka_publisher/    # Kafka producer with retry logic
│   │
│   ├── threat-intelligence/    # TIE analysis engine
│   │   ├── enrichment/
│   │   │   ├── ip_enricher/    # GeoIP, AbuseIPDB, Shodan
│   │   │   └── ioc_matcher/    # YARA + Suricata rule execution
│   │   ├── classifiers/
│   │   │   ├── ttp_mapper/     # MITRE ATT&CK TTP mapping
│   │   │   └── campaign_cluster/ # ML campaign clustering
│   │   ├── sandbox/            # CAPEv2 submission client
│   │   └── stix_exporter/      # STIX 2.1 / TAXII 2.1 server
│   │
│   ├── gateway/                # Network-layer interception
│   │   ├── dns_deceiver/       # DNS response manipulation
│   │   ├── traffic_redirector/ # iptables/nftables redirect rules
│   │   └── canary_monitor/     # Canary token callback receiver
```

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

```
│   │
│   ├── dashboard-api/          # REST + WebSocket API (FastAPI)
│   │   ├── routers/            # API route handlers
│   │   ├── auth/               # OAuth2/OIDC, RBAC
│   │   ├── websocket/          # Real-time event streaming
│   │   └── models/             # SQLAlchemy ORM models
│   │
│   └── dashboard-ui/           # React SPA
│       ├── src/
│       │   ├── views/
│       │   │   ├── WorldMap/
│       │   │   ├── SessionMonitor/
│       │   │   ├── ThreatFeed/
│       │   │   ├── Analytics/
│       │   │   └── AlertQueue/
│       │   ├── components/
│       │   ├── hooks/
│       │   └── store/          # Zustand state management
│       └── public/
│
├── infrastructure/
│   ├── docker/                 # Docker Compose (development)
│   ├── helm/                   # Kubernetes Helm chart
│   │   └── the-hook/
│   │       ├── charts/         # Sub-charts per service
│   │       ├── templates/
│   │       └── values.yaml
│   ├── terraform/              # Cloud infrastructure (AWS/GCP)
│   └── ansible/                # On-premise deployment playbooks
│
├── docs/                       # Architecture, runbooks, API docs
├── tests/                      # Integration and E2E test suites
└── scripts/                    # Utility scripts (setup, demo, seed)
```

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 11 — TECHNOLOGY STACK

11.1 Backend Services

Technology	Rationale
Python 3.12	Primary language for deception services and TIE. Rich ecosystem for security tooling (Scapy, Impacket, Paramiko, Volatility). Rapid development velocity for honeypot protocol implementation.
Go 1.22	High-performance networking components: gateway layer, telemetry collectors, Kafka producers. Go's concurrency model (goroutines) handles thousands of simultaneous attacker connections efficiently.
FastAPI (Python)	REST API and WebSocket server for the dashboard backend. Async-first, automatic OpenAPI documentation, strong typing with Pydantic. Production-grade performance with uvicorn.
Apache Kafka	Central event streaming backbone. Provides durable, ordered, replayable event log for all telemetry. Consumer groups enable independent processing by TIE, storage, and real-time dashboard.

11.2 Data Storage

Technology	Rationale
PostgreSQL 16 + TimescaleDB	Primary relational store for attacker profiles, alert state, and configuration. TimescaleDB extension provides native time-series optimisation for metrics and event counts.
Elasticsearch 8.x	Full-text search and aggregation over structured event data. Kibana (optional) for analyst ad-hoc querying. Standard in SOC toolchains, reducing integration friction.
Redis 7.x	Real-time state: active session tracking, rate

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.
THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

	limiting, pub/sub for WebSocket event fanout. Sorted sets for leaderboard-style attacker activity ranking.
MinIO (S3-compatible)	Object storage for session recordings, packet captures, and malware samples. S3-compatible API enables easy migration to AWS S3 in cloud deployments.

11.3 Infrastructure and Orchestration

Technology	Rationale
Kubernetes 1.30	Container orchestration for Hook Node lifecycle management. CRDs enable declarative deception topology definition. HPA provides automatic scaling of high-interaction nodes under load.
Helm 3	Package management for Kubernetes deployments. Enables parameterised deployment across cloud and on-premise environments with a single chart.
Terraform	Infrastructure-as-code for cloud resource provisioning (VPCs, security groups, EKS/GKE clusters). Reproducible, version-controlled infrastructure.
Cilium (eBPF networking)	Kubernetes CNI providing network policy enforcement and eBPF-based telemetry. Enables pod-level network visibility without performance overhead.

11.4 Frontend

Technology	Rationale
React 18 + TypeScript	Component-based SPA architecture. TypeScript eliminates a class of runtime errors in a complex state-heavy application.
D3.js + deck.gl	World map visualisation (deck.gl for WebGL-accelerated rendering), attack vector animations,

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.
THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

	ATT&CK heatmaps.
Zustand	Lightweight state management. Simpler than Redux for the dashboard use case; compatible with React 18 concurrent features.
Vite	Build tooling. Significantly faster hot-module replacement than webpack for developer experience.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 12 — CLOUD DEPLOYMENT

12.1 AWS Reference Architecture

The primary cloud deployment target is AWS. The following architecture uses managed services to minimise operational overhead while maintaining enterprise-grade security and availability.

AWS Service	Role	Notes
EKS (Kubernetes)	Hook Node orchestration, all microservices	Multi-AZ node groups, Spot instances for Hook Nodes
VPC with isolated subnets	Network isolation between deception env and management	Deception VLAN in private subnet, no internet egress
Amazon MSK (Kafka)	Managed event streaming backbone	3-broker cluster, TLS encryption in transit
RDS PostgreSQL	Primary relational database	Multi-AZ, automated backups, encryption at rest
Amazon OpenSearch	Elasticsearch-compatible event store	Fine-grained access control, UltraWarm for cold storage
ElastiCache Redis	Real-time state and pub/sub	Cluster mode, TLS, AUTH token
S3 + KMS	Encrypted object storage	Separate bucket per customer, KMS CMK per tenant
AWS WAF + Shield	Dashboard UI protection	Protects management endpoints from DDoS
CloudTrail + GuardDuty	Platform security monitoring	Detects unauthorised access to THE HOOK itself

12.2 Multi-Tenancy Architecture

THE HOOK supports three tenancy models to address different customer requirements:

- Shared SaaS: All customers share a single Kubernetes cluster. Customer data is isolated using Kubernetes namespaces, RBAC, and network policies. Appropriate for mid-market customers with standard compliance requirements.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

- Dedicated cluster: A dedicated EKS cluster per customer, deployed in THE HOOK's cloud account. Complete infrastructure isolation. Appropriate for enterprise customers with data residency requirements.
- Customer-managed (BYOC): THE HOOK Helm chart deployed in the customer's own cloud account. THE HOOK provides management plane connectivity for centralised monitoring and updates. Appropriate for regulated industries and sovereign cloud requirements.

12.3 Scalability Design

Hook Node scalability is achieved through Kubernetes Horizontal Pod Autoscaling triggered by active session count metrics, published to Prometheus. At 80% session capacity on existing nodes, new nodes are spawned automatically from the pre-built container image library. The system is designed to support thousands of concurrent attacker sessions without performance degradation — each Hook Node service is stateless and horizontally scalable.

The Kafka event pipeline is designed for linear throughput scaling: additional partitions and consumer group members can be added without downtime. At projected MVP scale (100 customers, 10,000 events/second peak), a 3-broker MSK cluster with m5.xlarge instances provides sufficient capacity. At 10× scale, this expands to a 9-broker cluster with no architectural changes.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 13 — SECURITY HARDENING

13.1 Platform Self-Protection Philosophy

THE HOOK operates in a uniquely adversarial environment — by design, it attracts sophisticated attackers. The platform must therefore be hardened against the possibility that an attacker inside the deception environment discovers they are in a honeypot and attempts to break out into the management infrastructure.

13.2 Sandbox Isolation

- Hardware-level isolation: High-interaction Hook Nodes run inside KVM virtual machines rather than plain containers. VM escape attacks against KVM are significantly more complex than container escape attacks.
- gVisor (container sandboxing): All container-based Hook Nodes run under gVisor (Google's user-space kernel). gVisor intercepts all system calls at the user-space boundary, preventing direct kernel exploitation.
- Network policy enforcement: Cilium network policies enforce strict communication rules — Hook Nodes can only communicate outbound to the Telemetry Collector. All other outbound connections are blocked at the eBPF layer.
- Read-only root filesystems: All Hook Node containers run with read-only root filesystems. Any filesystem modification by an attacker (malware installation, persistence mechanisms) occurs in an ephemeral writable layer that is purged when the session ends.
- No production network routing: The deception VLAN has no routing table entries pointing to production IP ranges. A physical ACL on the managed switch enforces this at the hardware level, independent of software configuration.

13.3 Management Plane Security

- mTLS everywhere: All inter-service communication within the Kubernetes cluster uses mutual TLS (cert-manager + Vault PKI). There are no plaintext internal channels.
- Secrets management: HashiCorp Vault manages all secrets (database credentials, API keys, encryption keys). No secrets are stored in Kubernetes secrets objects or environment variables directly.
- Zero-trust network access: The dashboard and management APIs are not accessible from the public internet. Access is via a Zero Trust Network Access (ZTNA) gateway (Cloudflare Access or Tailscale) requiring device certificate + identity provider authentication.
- Privileged access workstations: Platform engineers access production infrastructure only from hardened PAWs with full session recording and just-in-time privilege elevation.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

13.4 Data Protection

- Encryption at rest: All data stores (PostgreSQL, Elasticsearch, S3) use AES-256 encryption at rest with customer-managed KMS keys in dedicated cluster deployments.
- Data classification and minimisation: Captured attacker data is classified and handled according to its content. PII captured incidentally (e.g., in malware payloads) is flagged for legal review before any sharing or publication.
- GDPR/data protection compliance: THE HOOK's intelligence sharing features include automatic anonymisation of customer-identifying information. Shared threat intelligence contains only attacker-side indicators (IPs, hashes, TTPs) — never customer infrastructure details.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 14 — PROOF OF CONCEPT DEMO

14.1 Demo Scenario: The Corporate Infiltrator

The following scenario is designed for a 20-minute investor or enterprise prospect demonstration. It requires a pre-configured demo environment and a second laptop or mobile device acting as the 'attacker'.

Step	Attacker Action	THE HOOK Response
1. Reconnaissance	Runs nmap scan against demo network range	Gateway detects scan; Hook Nodes respond with realistic banners; dashboard shows incoming scan with attacker IP geolocation
2. Service Discovery	Discovers fake SSH server (22), HTTP server (443), MySQL (3306)	All three probe events logged; TIE enriches attacker IP; alert fires: 'New attacker session initiated'
3. Credential Attack	Attempts SSH bruteforce with common password list	FakeSSH logs 47 credential attempts; dashboard displays real-time credential feed; after threshold, accepts login
4. Exploration	Runs whoami, ls, finds /home/admin/finance/Q4_Budget.xlsx	Every command keystroke logged; file access event triggers 'Sensitive File Access' alert
5. Lateral Movement	Discovers fake database credentials in .env file; connects to MySQL	MySQL honeypot accepts connection; attacker queries fake customer table; 'Data Exfiltration Attempt' alert fires
6. Malware Drop	Uploads reverse shell script via SCP	Malware auto-submitted to analysis pipeline; YARA match identifies Netcat-based reverse shell; full static report generated in <90 seconds
7. Intelligence Report	Demo ends	Dashboard shows complete session timeline, ATT&CK TTP heatmap, enriched attacker profile, and exportable STIX bundle

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

14.2 Demo Setup Requirements

- Demo environment: Docker Compose deployment on a MacBook Pro M3 or equivalent. Full environment spins up in under 5 minutes.
- Pre-seeded data: 72 hours of pre-recorded attacker sessions provide a realistic-looking historical dashboard state for the analytics views.
- Attacker laptop: Second device running Kali Linux or a simple bash script that triggers the scan and connection sequence automatically for a smooth demo flow.
- Slide deck integration: Dashboard walkthrough projected alongside the attack simulation, narrated by the presenter with pre-prepared talking points for each stage.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 15 — STARTUP PRODUCTISATION

15.1 From MVP to Commercial Product

The transition from MVP to a commercial-grade product requires systematic work across engineering maturity, compliance readiness, operational tooling, and go-to-market packaging. The following roadmap spans 18 months post-MVP.

Release	Timeline	Key Additions
V1.0 (MVP)	Month 0	3 Hook Node profiles, basic TIE, React dashboard, Docker Compose deployment
V1.1	Month 3	Kubernetes Helm chart, 10 Node profiles, Slack/PagerDuty integration, basic API
V1.2	Month 6	Multi-tenancy, SAML SSO, SIEM connectors (Splunk, Sentinel), SLA-backed cloud offering
V2.0	Month 9	Adaptive Environment Engine, ML campaign clustering, STIX/TAXII export, MSSP portal
V2.1	Month 12	Full ATT&CK coverage, IoT/OT profiles, canary token platform, automated threat reports
V3.0	Month 18	AI-driven attacker engagement, predictive intelligence, federated threat sharing network

15.2 SaaS Platform Model

The SaaS deployment model targets organisations that want immediate time-to-value without infrastructure investment. THE HOOK SaaS provides a fully managed deception environment, deployed as a virtual extension of the customer's network via a lightweight gateway appliance (software or hardware) installed on-premise.

The SaaS model requires THE HOOK to operate as a multi-tenant cloud service with strict data isolation, 99.9% uptime SLAs, and a self-service onboarding experience. Customers configure their

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

deception topology through the dashboard — no professional services engagement required for standard deployments.

15.3 Enterprise Deployment Model

Enterprise customers with complex compliance requirements (financial services, defence, critical infrastructure) are served through a dedicated deployment model: a single-tenant THE HOOK instance deployed in the customer's cloud account or on-premise data centre, managed by THE HOOK's SRE team under a managed service agreement.

15.4 MSSP White-Label Programme

Managed Security Service Providers (MSSPs) represent a high-leverage distribution channel. THE HOOK offers a white-label programme enabling MSSPs to rebrand and resell the platform under their own identity. The MSSP portal provides multi-customer management capabilities, consolidated billing, and a partner-tier API for integration with MSSP SOC platforms.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 16 — BUSINESS MODEL

16.A Enterprise Cybersecurity Product (Licence Model)

The enterprise licence model is appropriate for large organisations (>1,000 employees) with existing security infrastructure and compliance requirements that preclude cloud data sharing.

Tier	Description / Pricing
Hook Starter (up to 50 nodes)	€18,000/year — Up to 50 Hook Nodes, 5 protocol emulators, basic TIE, email alerting, 1 dashboard user seat
Hook Professional (up to 200 nodes)	€48,000/year — Up to 200 nodes, full protocol library, ML TIE, SIEM integration, 5 user seats, priority support
Hook Enterprise (unlimited)	€120,000+/year — Unlimited nodes, Adaptive Engine, STIX export, dedicated account manager, custom SLA, 25+ seats
Professional Services	€1,800/day — Architecture review, custom profile development, red team validation engagements
Annual Threat Intelligence Report	€8,000/year — Customer-specific annual intelligence report based on captured attacker data

16.B Hook as a Service (Subscription Model)

The SaaS subscription model targets mid-market organisations (100–1,000 employees) that lack the internal resources to manage a complex security product. All infrastructure is managed by THE HOOK; the customer pays a monthly subscription.

Plan	Pricing / Features
Starter	€990/month — 10 Hook Nodes, 3 protocols, email alerts, standard dashboard, community threat feed
Professional	€2,490/month — 50 nodes, full protocols, SIEM integration, Slack/Teams alerts, priority TIE, 99.9% SLA

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

Business	€5,990/month — 150 nodes, Adaptive Engine, dedicated IP enrichment budget, STIX export, quarterly intel briefing
MSSP Partner	Volume pricing — Per-endpoint licensing for resale; minimum 500-endpoint commitment; white-label dashboard

16.C Revenue Projections

Year	Target Customers	ARR Estimate
Year 1 (MVP + early sales)	5–10 pilot customers (mix of SaaS + enterprise)	€150,000 – €400,000
Year 2 (product-market fit)	30–60 customers	€1,500,000 – €3,500,000
Year 3 (scale)	100–200 customers, MSSP channel active	€5,000,000 – €12,000,000

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 17 — GO-TO-MARKET STRATEGY

17.1 Phase 1 — Pilot Customer Acquisition (Months 1–6)

The first six months focus entirely on acquiring three to five design partner / pilot customers who are willing to deploy the MVP in exchange for significantly discounted or free access and the commitment to provide structured feedback.

- Target profile: CISOs at mid-size financial services, fintech, or SaaS companies with existing security teams. These organisations have the technical sophistication to engage meaningfully with the platform and the budget sensitivity to appreciate a value-first commercial approach.
- Acquisition channel: Founder-led outreach via LinkedIn and personal network. Direct pitch to CISOs and security architects at target companies. Offer: 90-day free pilot with full support; no obligation to purchase.
- Success criteria: Each pilot customer generates at least one documented case study demonstrating: attacker activity detected within 30 days, intelligence value generated (TTPs, IOCs, attacker profiles), and measurable improvement in security posture.

17.2 Phase 2 — Market Presence (Months 6–12)

- Security conferences: Presence at Black Hat Europe, x33fcon, Hack In The Box, and regional BSides events. Live demo booth with THE HOOK demo scenario running continuously. Speaking submissions to practitioner tracks.
- Content marketing: Technical blog posts on honeypot strategy, case studies (anonymised), open-source release of specific non-core components (e.g., protocol emulation libraries) to build developer community credibility.
- Analyst outreach: Briefings with Gartner (Deception Technology market coverage), Forrester, and IDC. Target inclusion in relevant Market Guides and Emerging Technology reports.
- MSSP partnerships: Identify 2–3 European MSSPs as white-label partners. MSSPs provide instant access to their managed customer base without direct sales cost.

17.3 Phase 3 — Scaling (Year 2+)

- Inside sales team: Hire 2–3 enterprise account executives targeting the €48K+ professional tier.
- Channel partner programme: Formalise MSSP and VAR (Value-Added Reseller) programme with deal registration, co-marketing funds, and partner certification.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

- International expansion: Prioritise DACH region (Germany, Austria, Switzerland) given strong cybersecurity investment culture and regulatory environment (NIS2, DORA). Secondary target: Benelux and Nordics.
- Threat intelligence product: Launch standalone threat intelligence subscription leveraging network-effect data from the growing customer base. This becomes a second revenue stream and a market differentiation story.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

SECTION 18 — LONG-TERM VISION

18.1 The Intelligence Network Effect

The most compelling long-term value of THE HOOK lies not in any individual deployment but in the aggregate intelligence generated across the entire customer network. As the platform scales to hundreds and then thousands of deployments, each customer deployment contributes attacker observations that improve detection and response for all others.

This creates a compounding competitive advantage that is structurally difficult for a new entrant to replicate. A competitor that launches a technically equivalent product with zero customers will generate zero network intelligence — they cannot match the breadth and depth of THE HOOK's threat picture regardless of their individual product quality.

VISION	At 500+ enterprise deployments, THE HOOK becomes one of the world's largest attacker observation networks — a strategic intelligence asset with value extending far beyond individual enterprise security deployments, into national cybersecurity infrastructure and international threat intelligence sharing ecosystems.

18.2 AI-Driven Attacker Engagement

The next generation of the Adaptive Environment Engine will incorporate large language model (LLM) capabilities to conduct genuinely interactive, contextually aware conversations with attackers operating in the deception environment. Rather than static filesystem content, an LLM-powered fake sysadmin persona can respond to chat messages, answer questions plausibly, and adapt the environment narrative in real time.

This dramatically increases attacker dwell time and intelligence extraction potential — particularly against sophisticated human operators (as opposed to automated tools) who would immediately detect a static honeypot.

18.3 Platform Expansion Directions

- Cloud-native deception: As enterprise infrastructure migrates to cloud, THE HOOK expands its deception coverage into cloud-native attack surfaces — fake S3 buckets, EC2 metadata service traps, Kubernetes API server honeypots, and serverless function monitoring.
- Supply chain deception: Integration with software supply chain security tooling — fake npm packages, PyPI packages, and container images that beacon home when installed, enabling detection of supply-chain compromise campaigns.
- Active threat hunting: Evolution from passive observation to active threat hunting — using intelligence from the deception environment to proactively identify compromised assets in the production network.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

- Regulatory and compliance intelligence: Automatic mapping of observed attacker TTPs to regulatory control frameworks (NIS2, ISO 27001, SOC 2) to provide evidence of threat-informed control validation.

18.4 Strategic Exit Paths

THE HOOK's strategic positioning makes it an attractive acquisition target for multiple buyer categories within a 5–7 year horizon:

- Tier-1 security vendors: CrowdStrike, Palo Alto Networks, and SentinelOne have all acquired deception technology capabilities historically (Illusive Networks, Attivo Networks). THE HOOK's network intelligence asset and customer base would complement their existing platforms.
- Managed security service providers: Large MSSPs (SecureWorks, Trustwave, Orange Cyberdefense) seeking to differentiate their managed detection and response offerings with proprietary deception capabilities.
- Threat intelligence platforms: Recorded Future, Flashpoint, and Mandiant (Google) operate large threat intelligence businesses. THE HOOK's attacker observation network would provide a unique, live, first-party intelligence source.

At €5M ARR with strong growth, the platform warrants a 10–15× revenue multiple — a transaction value of €50M–€75M — representing a compelling return on initial investment for both founders and Series A investors.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

APPENDIX — KEY DEFINITIONS & REFERENCES

Term	Definition
Honeypot	A decoy system or service designed to attract attackers and observe their behaviour.
Honeynet	A network of interconnected honeypots forming a simulated production environment.
TTPs	Tactics, Techniques, and Procedures — the specific methods used by a threat actor.
MITRE ATT&CK	A globally accessible knowledge base of adversary tactics and techniques based on real-world observations.
IOC	Indicator of Compromise — forensic evidence that a system has been breached (e.g., IP addresses, file hashes).
STIX / TAXII	Structured Threat Information eXpression / Trusted Automated eXchange of Intelligence Information — standard formats for threat intelligence sharing.
Canary Token	A trap credential or URL that, when used, generates an immediate alert indicating theft or use by an attacker.
Dwell Time	The period between an attacker's initial compromise and detection by the defender.
TTP Mapping	The process of categorising observed attacker behaviour against the MITRE ATT&CK framework.
MSSP	Managed Security Service Provider — a third-party company providing outsourced security monitoring and management.
ECS	Elastic Common Schema — a standardised event schema for normalising security events.
eBPF	Extended Berkeley Packet Filter — a kernel technology enabling high-performance network and system monitoring.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK — Confidential Technical Architecture Document

Built with precision. Designed to deceive.

 © 2026 Ciprian Stefan Plesca. All Rights Reserved.

THE HOOK™ and all associated materials are proprietary intellectual property. Unauthorized copying, reproduction, modification, distribution, public reposting, derivative commercialization, or operational deployment is strictly prohibited without written permission.

© 2026 Ciprian Stefan Plesca. All Rights Reserved.